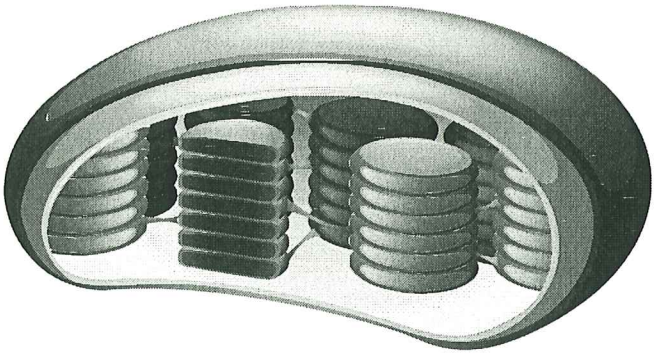
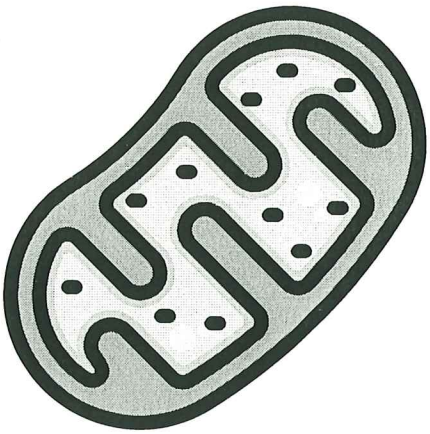
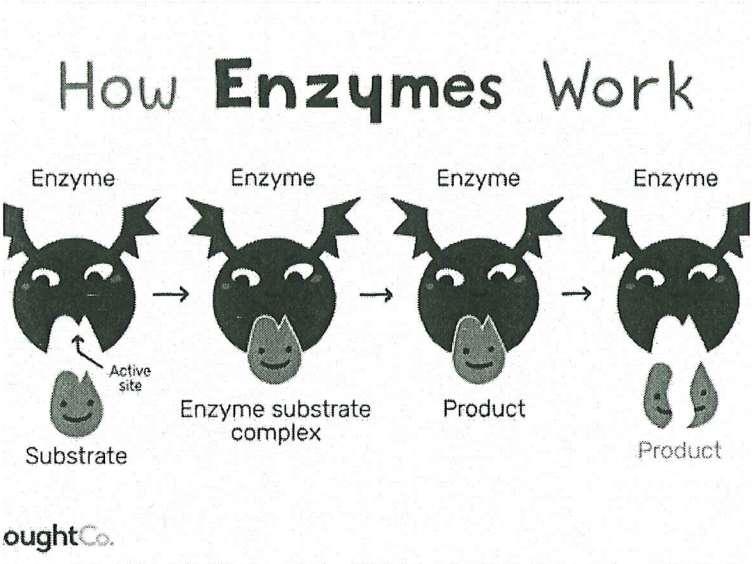
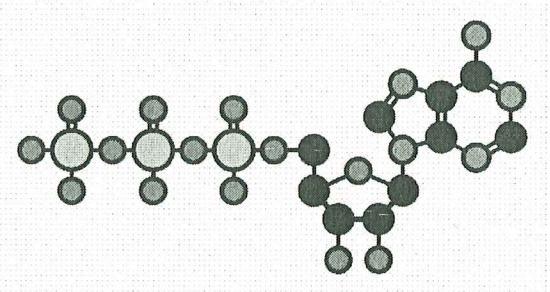
Unit 3: Cellular Energetics









Topics:

Enzymes

ATP

Cellular Respiration

Photosynthesis

Metabolism and Enzymes

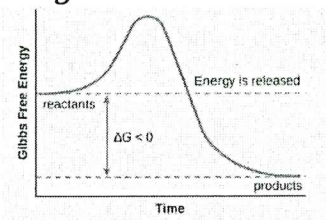
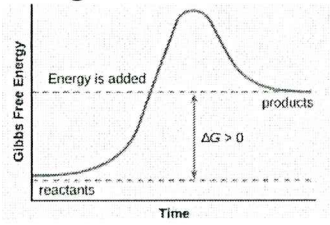
Metabolism vs. Catabolism

Metabolism	Catabolism

Chemical Reactions

Law of conservation of matter:
What happens to atoms during a chemical reaction?

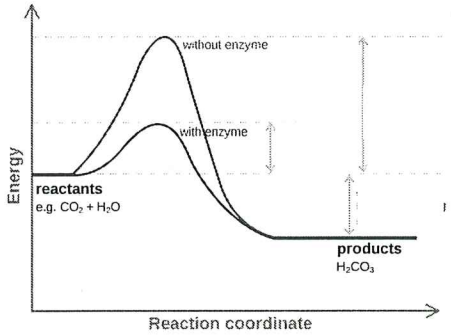
Energy in Reactions

Exergonic Reaction	Endergonic Reaction
	

Enzymes

Activation Energy	Catalyst	Enzymes

Label the enzyme graph:



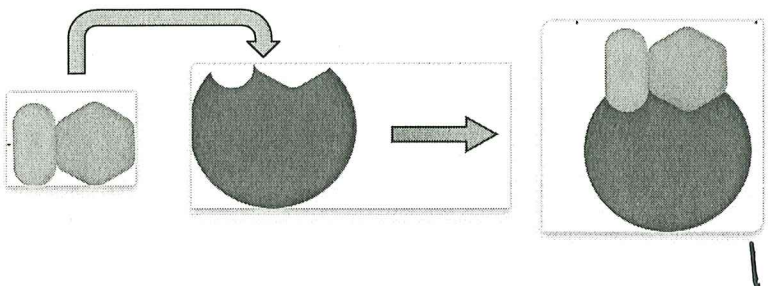
Enzyme Substrate Complex

Substrate:

Active Site:

Enzyme Substrate Complex:

Label the enzyme complex:

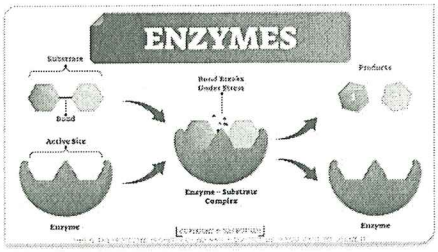


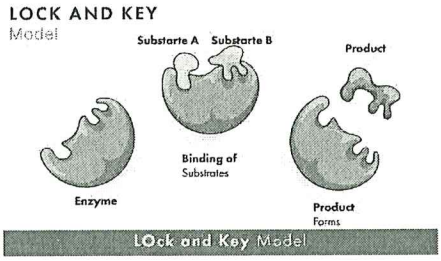
Lock and Key Mechanism

How can enzymes be compared to a **lock and key**?

What happens to enzymes after the reaction is complete?

Enzymes can catalyze anabolic or anabolic reactions

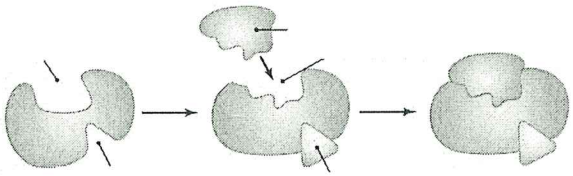




Allosteric Activation

- Allosteric Enzymes:

- Effector Molecules:



Allosteric activation

Changes to Enzyme Shape:

- Factors that affect shape:
- Denaturation:

How would denaturation affect a protein's function?

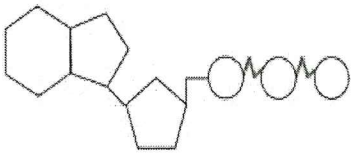
Inhibition

Non-Competitive Inhibition	Competitive Inhibition

ATP and Energy

ATP

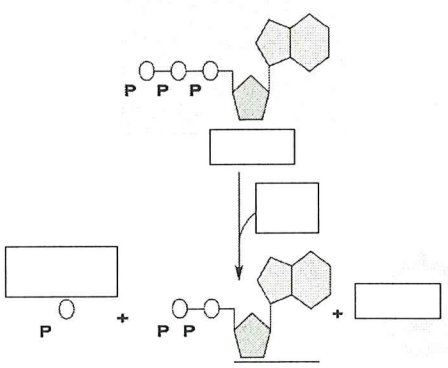
Label the ATP molecule below



Energy for the Cell

Which bond of ATP has the most stored energy?

ATP Hydrolysis:

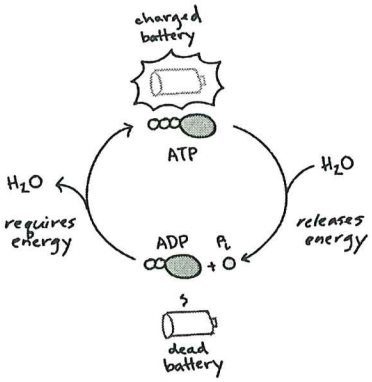


Is ATP Hydrolysis catabolic or anabolic? Exergonic or endergonic?

What happens to the phosphate group?

Phosphorylation:

ATP ADP Cycle:



How can we relate ATP/ADP to batteries?

Harvesting Stored Energy:

How do we harvest energy from fuels?

Oxidation and Reduction

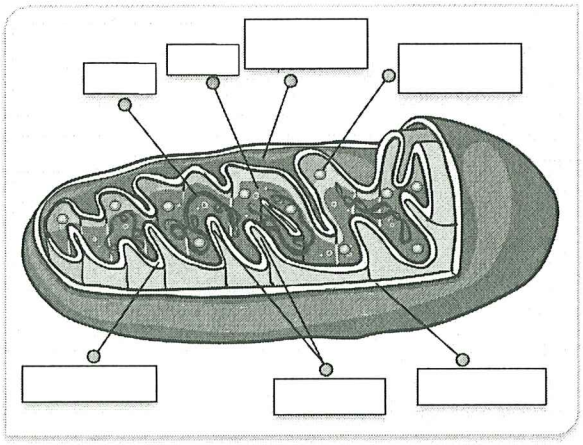
OIL RIG

Cellular Respiration

Overview- Cell Respiration
Goal:
Reactants:
Products:

Equation for Cell Respiration:

Structure of the Mitochondria



- Double membrane:
- Inner membrane:
- Cristae:
- Intermembrane space:
- Matrix:
- What makes the mitochondria unique from other organelles?

Overview- Steps to Cellular Respiration

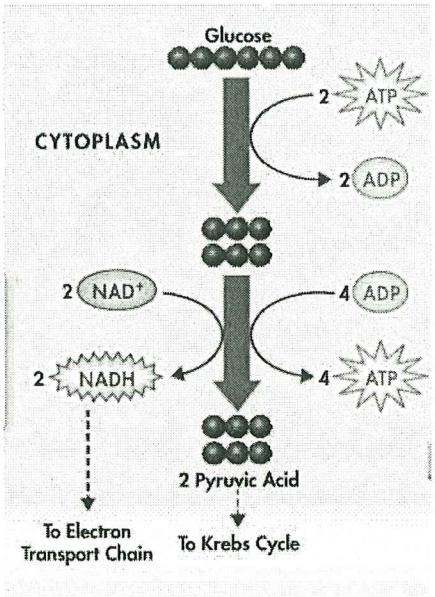
- 4 Metabolic Stages
- Anaerobic Respiration
- Aerobic Respiration

- Gradual _____ of _____ releases the energy to make _____ at each step.
- _____ and _____ are carried down to the last step (Electron Transport Chain)
- _____ and _____ are coenzymes that act as _____ for the cell and shuttle electrons to the ETC
  - NAD+ is _____ into _____ and FAD is _____ into $FADH_2$

Step 1: Glycolysis

Glycolysis Overview

- Glycolysis:
- Location:
- Type of respiration:
- How much ATP is produced?



Evolutionary Perspective: Why is glycolysis anaerobic?

Steps to glycolysis:

Step 1:

Step 2:

Step 3:

Summary of Glycolysis:

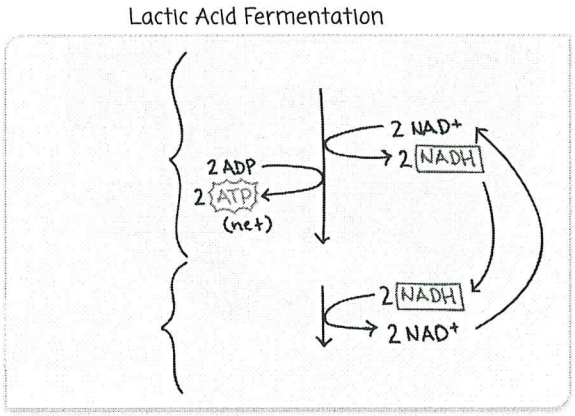
- Converts:
- Produces:
- Consumes:
- Net Yield:
- Where does the NADH go?

Branching point:

- If oxygen is present:
- If no oxygen is present:

Lactic Acid Fermentation:

- Performed by:
- Pyruvate is converted to:
- Significance:
-
- Is it reversible?



Alcohol Fermentation

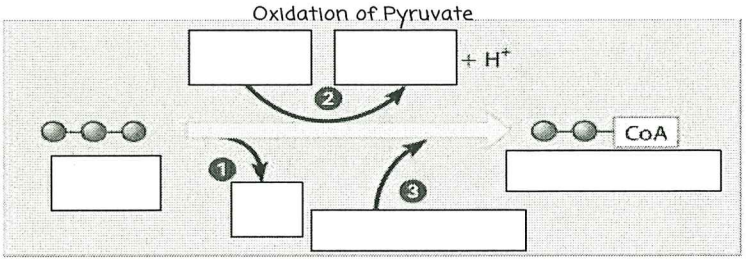
- Performed by:
- Used for:
- Pyruvate is converted to:
- Produces:
- Is it reversible?

What is the overall goal of BOTH types of fermentation?

Step 2: Oxidation of Pyruvate

Steps to Oxidation of Pyruvate

1.
2.
3.



What are the end products and where do they go?

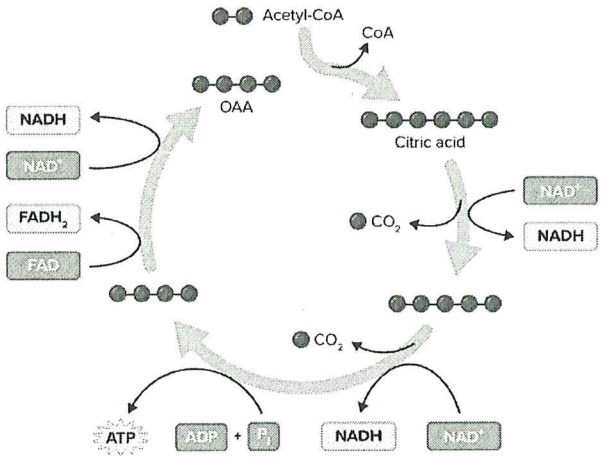
Step 3: The Krebs Cycle

Who identified this pathway and what is another name for it?

What happens in this pathway and what is the overall goal?

Stages of the Krebs Cycle

1. Formation of Citric Acid



2. Oxidation of Citric Acid

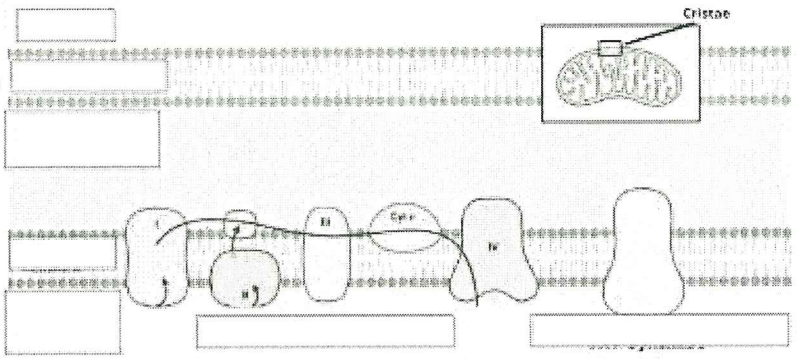
3. Regeneration of Oxalacetate

What are the results of the Krebs cycle and what happens to the electron carriers?

Step 4: The Electron Transport Chain

Summary of Reactions so Far:

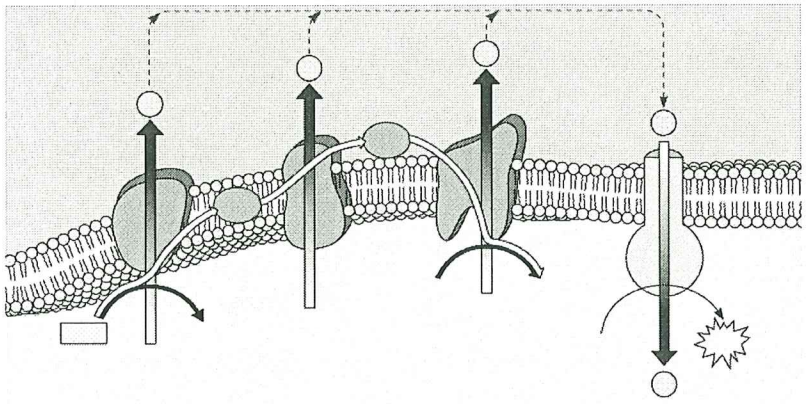
Structures of the ETC



How does the mitochondria make its own proteins?

ETC Stage 1: Building a Proton Gradient

Stage 2: Oxidative Phosphorylation



Copyright © Pearson Education, Inc., publishing as Benjamin Cummings.

What would happen if there was no oxygen during the ETC?

Cellular Respiration Review Questions:

Where did the glucose come from?

Where did the O2 come from?

Where did the CO2 come from?

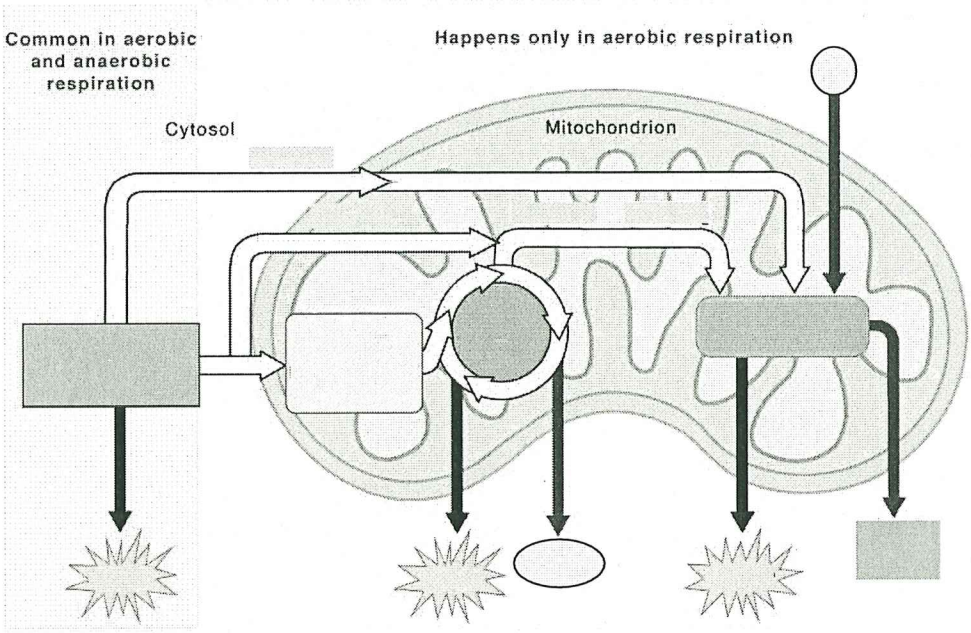
Where did the CO2 go?

Where did the H2O come from?

Where did the ATP come from?

What else is produced that is not listed in this equation?

Why do we breathe?



Photosynthesis

Autotrophs	Heterotrophs

Plant Anatomy Basics

- Roots:
- Stems:
- Leaves:
- Transpiration:

What do plants need and where do they get it?

Photosynthesis Equation:

How is this equation similar to the equation for cellular respiration?

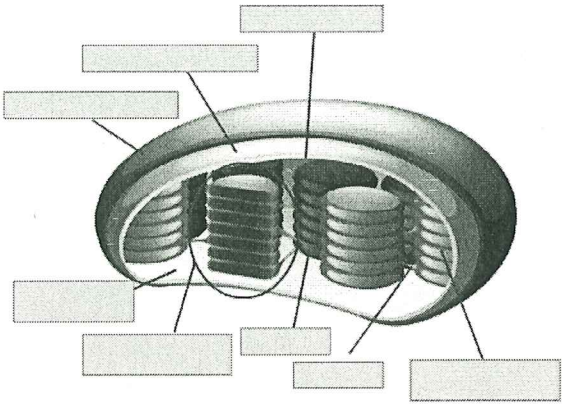
Chloroplasts

Photosynthesis location:

Two main Parts:

- Grana:
- Stroma:

Thylakoid Membrane:



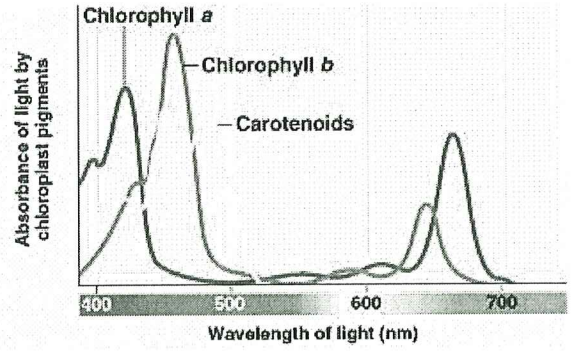
How is the chloroplast similar to the mitochondria?

Photosynthesis Overview

- Light Reactions:
- Calvin Cycle:

Light Absorption

Pigments of Photosynthesis



-Chlorophyll:

-Chlorophyll a:

-Acessory pigments:

-Which color of light does NOT get absorbed by plants?

Photosynthesis Step 1: Light Dependent Reactions

Light Reactions Overview
-Location:
-Summary:

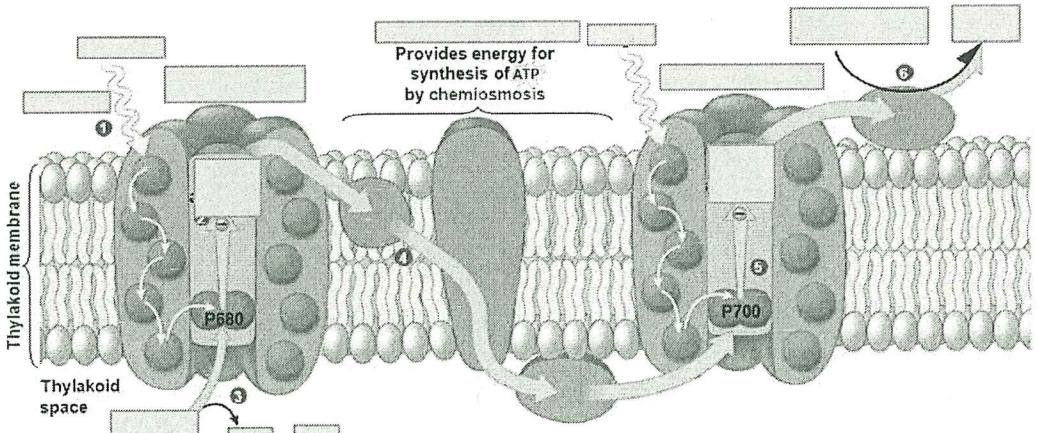
-Byproducts:
-Photosystems:

-ETC 1:
- Pairs with:
- Makes:

-ETC 2:
- Pairs with:
- Makes:

Step 1: Photosystem II/ETC 1

Step 2: Photosystem II/ETC 1



▪Electrons are

▪Between the two photosystems, the electrons

Photosynthesis Step 2: The Calvin Cycle

Review: What do plants need?

- Plants need:

- What was made in the light reactions?
- How do plants store chemical energy?
- Where is this energy stored?

How is Glucose Made?

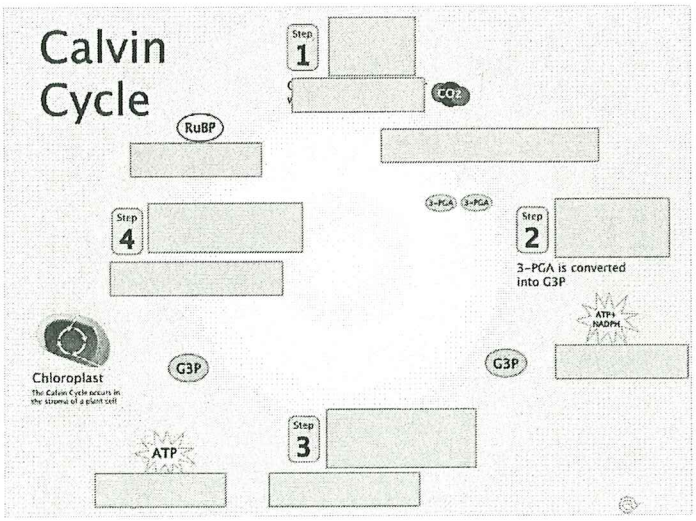
-What powers this reaction?

Calvin Cycle Overview
-Location:
-Powered by:
-Why can't we use CO_2 for energy?

-Synthesizing Glucose=

Calvin Cycle Overview

- Location:
- Powered by:
- CO2 is converted:
- Synthesizing glucose:



Calvin Cycle In Detail

Major Phases:

Carbon Fixing Enzyme

Calvin Cycle Summary

Summarizing Photosynthesis

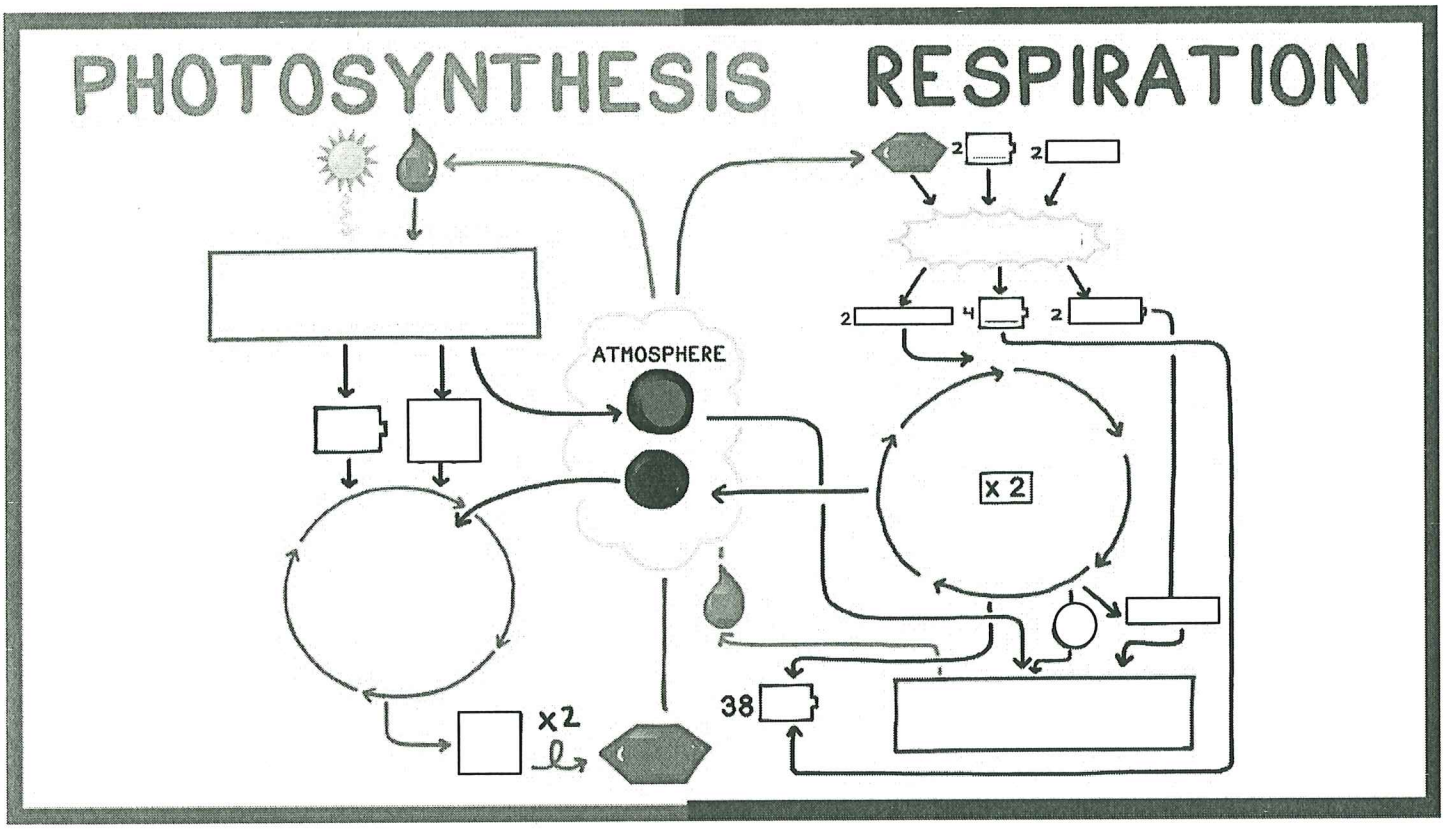
Light Reactions	**Calvin Cycle**
Consumed:	Consumed:
Produced:	Produced:
Released:	Regenerated:

Summarize the way in which producers and consumers support each other.

Supporting a Biosphere

Use the following terms to relate photosynthesis to cellular respiration (some will be used more than once)

H2O	NADH2	Calvin Cycle	ETC
O2	NADPH	LIght Reactions	Glycolysis
CO2	G3P	Pyruvate	Kreb's Cycle
FADH2	Glucose	NAD+	



Name:_____ Period:_____

Enzyme Practice + Cut and Paste

Background: Enzymes are proteins that help chemical reactions by acting as a catalyst. A catalyst lowers the activation energy needed to start a reaction. Enzymes first have to react with a substrate. Each enzyme has a specific shape that only fits with a specific substrate. Once the enzyme substrate complex is formed, the enzyme will catalyze the reaction and release a product.

Directions: You will be answered questions about enzymes and using molecule pieces to demonstrate how enzymes work. Molecules to cut and paste are located on the last page of this packet.

Part A- Vocabulary- Define the following:

Vocab Word	**Definition**
Enzyme	
Catalyst	
Chemical reaction	
Activation Energy	
Substrate	
Active Site	
Denaturation	

Part B: Demonstrate a Catalyzed Catabolic Reaction- Lactase

Lactase is the enzyme we will use to catalyze a catabolic reaction. Lactase catalyzes the reaction that breaks down lactose into glucose and galactose

Use the cutouts or draw your own on the last page to show the following: Enzyme + Substrate → Enzyme-substrate complex → Enzyme + Product 1 + Product 2

Label the following: Enzyme, substrate, enzyme substrate complex, active site, products

Part C: Demonstrate a Catalyzed Anabolic Reaction- Sucrase

Sucrase can catalyze an anabolic reaction that combines glucose and fructose into sucrose.

Use the cutouts or draw your own on the last page to show the following: Enzyme + Substrate 1 + substrate 2 → Enzyme-substrate complex → Enzyme + Product

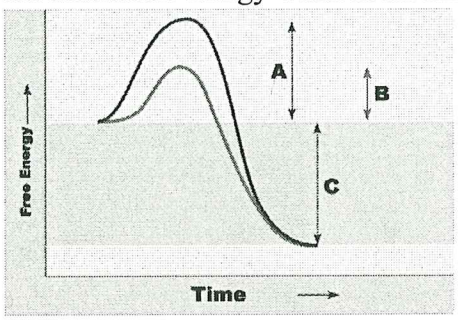
Label the following: Enzyme, substrate, enzyme substrate complex, active site, products

Part D: Practice Questions

1. What type of macromolecule are enzymes?
2. Why is the active site important in enzymes?

3. Under certain conditions enzymes change shape and are no longer functional. What is this shape change called?

4. In the graph below, label the activation energy without an enzyme and the activation energy with an enzyme.

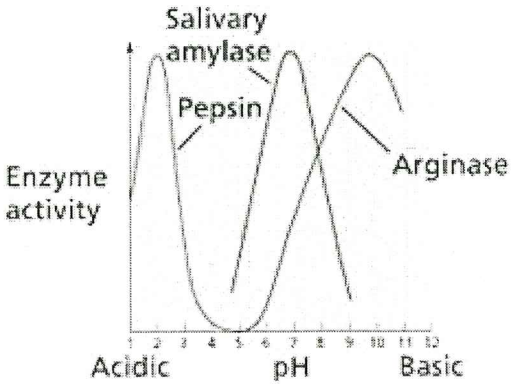


5. The graph to the right shows the activity of three different enzymes.

   a. Which enzyme works best in acidic conditions?

   b. Which enzyme works best in alkali conditions?

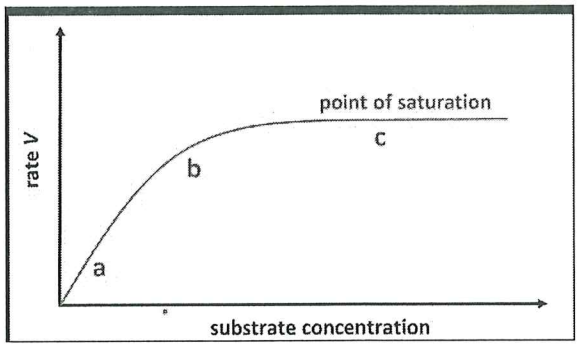
   c. Which enzyme would work best at pH of 7?



6. Use the graph to the right to answer these questions:
   a. Which point shows the enzyme increasing in activity?

   b. Which point represents the maximum amount of activity the enzyme will have?

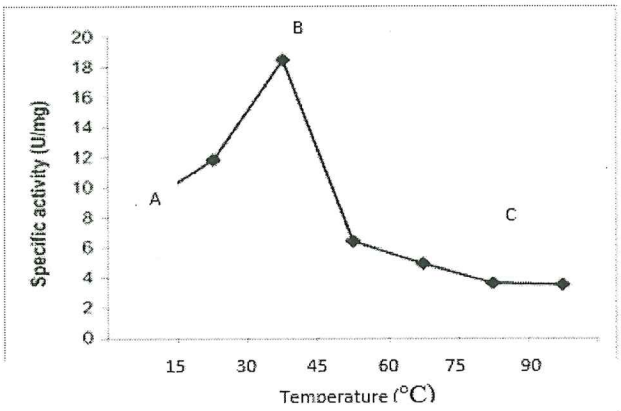
   c. Why won't the reaction rate increase past the point of saturation?



7. The graph below shows how temperature affects an enzyme-controlled reaction. Study the graph and answer the questions below.

   a. What is happening between point A and B?

   b. What does point B represent?

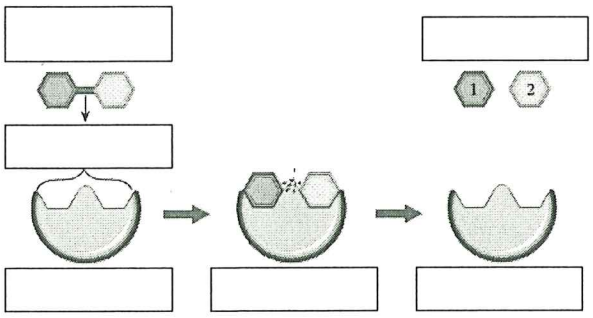


   c. Explain what is happening between point B and point C. What is causing this?

   d. What happens to the active site when a temperature is too high? Draw a picture.

Cellular Energy Review Stations

Station 1: Enzymes



Is this enzyme catalyzing a catabolic or anabolic reaction? How do you know?

Looking at the images below, how are competitive and non-competitive inhibition different?



Graph A Questions:

1. Which enzyme works best under acidic conditions?
2. Which enzyme works best under alkaline conditions?
3. Would salivary amylase still be able to function if it was exposed to an environment with a pH of 2? Why or why not?

Graph B Questions:

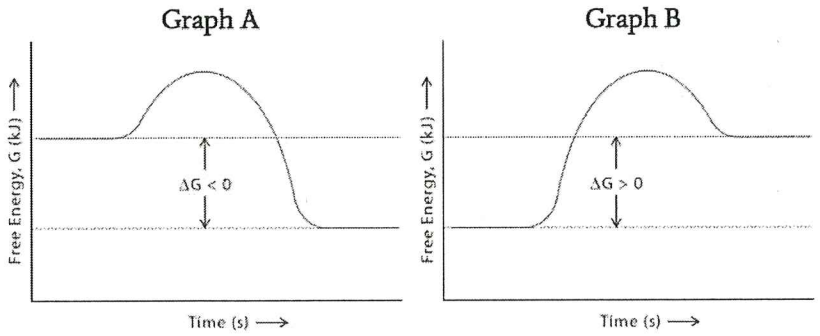
1. At what temperature does this enzyme work best?
2. What process is happening to the enzyme between point B and C?
3. What happens to the active site of an enzyme when the temperature is too high? Would the enzyme still be able to function? Why or why not?

Graph C Questions:

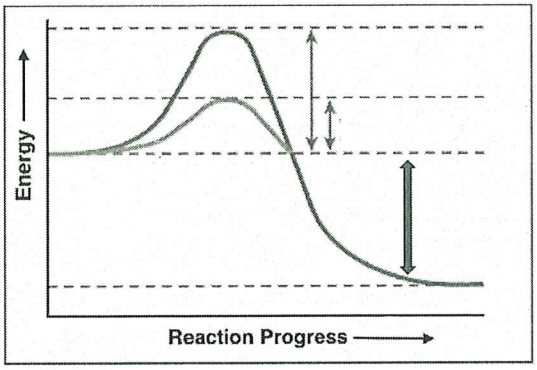
1. What is happening between point a and b?
2. What is happening at point c?
3. Why won't the reaction rate increase past the point of saturation?

Station 2: Energy

Label the parts of each graph below.



Label the parts of the graph below.



How do enzymes speed up the rate of chemical reactions?

Station 3: Relating Cellular Respiration and Photosynthesis

	Cellular Respiration	Photosynthesis
Role of Oxygen		
Role of CO2		
Role of Glucose		
Role of Water		
Role of ATP		
WHO does it?		
WHERE does it occur in the cell?		
WHY does it occur?		

How are cellular respiration and photosynthesis related processes? Include anabolic and catabolic in your explanation.

Station 4: Cellular Respiration

Chemical Reaction:			
	Location	Reactants	Products
Cellular Respiration (overview)			
Glycolysis			
Fermentation			
Oxidation of Pyruvate			
Kreb's cycle			
Electron Transport Chain			

1. What is the process that occurs after glycolysis, when no oxygen is available in animals, and in bacteria?

2. What is the stage of cellular respiration in which Acetyl-CoA is broken down to release carbon dioxide, and create, NADH and FADH2?
3. What is the process that releases energy by breaking down food molecules in the presence of oxygen?

4. Kearns-Sayre Syndrome is a mitochondrial disease in which the part of mitochondria responsible for carrying oxidative phosphorylation is damaged or does not function properly. What specific consequences does this disease have to the process of cellular respiration?

Station 5: Photosynthesis

Chemical Reaction:			
	Location	Reactants	Products
Photosynthesis (overview)			
Light Dependent Reactions			
Photosystem II			
Photosystem I			
Calvin Cycle			

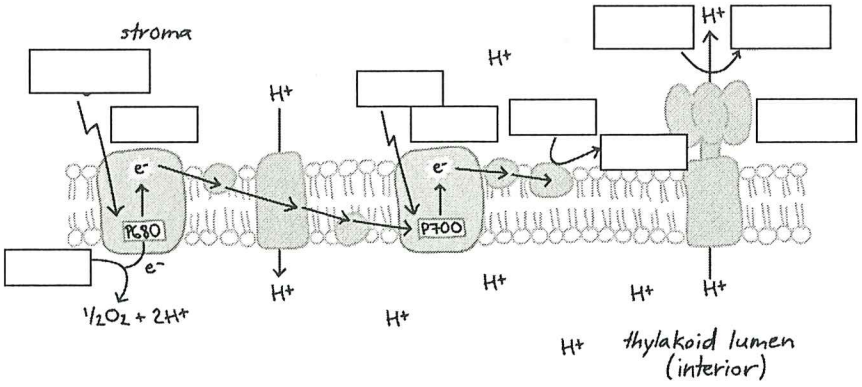
1. What is the process called by which plants use the sun's energy to make high-energy sugars?

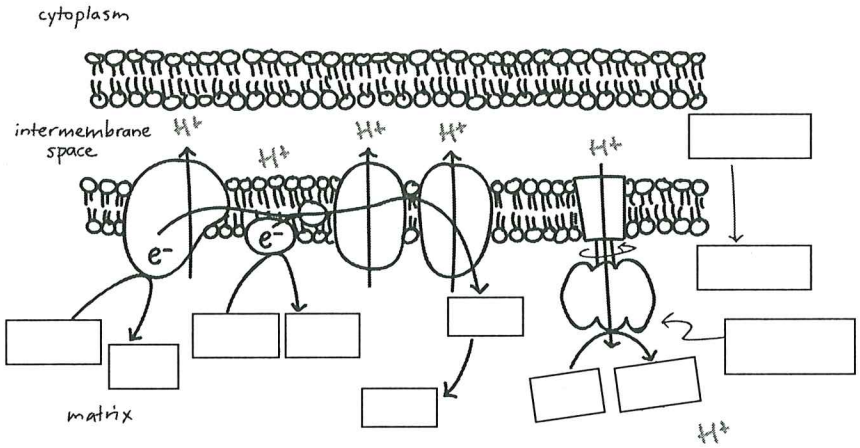
2. What is rubisco? Why is it important?

3. What is an organism called that makes its own food? Do these organisms also contain mitochondria? Why or why not?

4. How would the closing of stomata impact photosynthesis?

Station 6: The ETC





	Cellular Respiration	Photosynthesis
What is the purpose of the ETC?		
Who are the electron carriers?		
Where is the ETC located?		
What supplies the energy for the ETC?		

Station 7: CR Practice Questions

1. In most hypotheses of early Earth and the formation of life, it is accepted that the atmosphere was a reducing one. This means that there was an abundance of methane, ammonia, and water vapor, but there was an absence of significant amounts of oxygen gas. Based on this assumption, what was likely the first metabolic activity to harness ATP to evolve in primitive cells?
    a. Photosynthesis, as plants evolved before animals
    b. Cellular respiration, as this generates the most ATP
    c. Glycolysis, as this does not require organelles nor oxygen
    d. Calvin cycle, as glucose was required by all cells for metabolism

2. In which type of respiration are alcohol and lactic acid produced?
    a. Aerobic respiration
    b. Anaerobic respiration
    c. Photorespiration
    d. Both aerobic and anaerobic respiration

3. Which type of respiration is more efficient and why?
    a. Aerobic respiration because it produces a total of 2 ATP
    b. Anaerobic respiration because it produces a total of about 36 ATP
    c. Aerobic respiration because it produces 36 ATP
    d. Anaerobic respiration because it produces 2 ATP

4. Which of the following processes generates the most ATP?
    a. Electron transport chain
    b. Krebs cycle
    c. Glycolysis
    d. Fermentation

5. If there is no oxygen available after glycolysis, what is the next step in the respiration pathway?
    a. Krebs cycle
    b. ETC
    c. Fermentation
    d. Nothing will happen

6. _____ is diffused into the inner membrane space during the electron transport chain. _____ is formed as a byproduct when oxygen is reduced in the ETC.
    a. CO2; H2O
    b. O2; CO2
    c. H2O; O2
    d. O2; H2O

7. All of the following occur during the Krebs cycle EXCEPT:

    I. Acetyl-CoA combines with oxaloacetate to form citrate.
    II. FADH2 accepts two electrons to form FAD.
    III. NAD+ is reduced to form NADH.
    IV. Pyruvate is converted into acetyl-CoA
    V. Carbon dioxide, NADH, and FADH2, and ATP are reactant

    a. I and III
    b. II, IV, and V
    c. II, III, and IV
    d. I, II, IV, and V

8. Imagine there are two different cells, cell A and cell B. Explain why lactic acid, carbon dioxide and only 2 ATP are produced in cell A and carbon dioxide, water and up to 32 ATP are produced in cell B.
    a. cell A is in an animal cell in an anaerobic environment and cell B is an animal cell in an aerobic environment
    b. cell A is an animal cell in an aerobic environment and cell B is an animal cell in an anaerobic environment
    c. cell A is a yeast cell in an anaerobic environment and cell B is an animal cell in an aerobic environment
    d. cell A is an animal cell in an aerobic environment and cell B is a yeast cell in an aerobic environment

9. Heterotrophs take in mass in the form of food. Food is either used to build molecules within the body or as fuel for cellular respiration. If used for cellular respiration, this mass will eventually be converted into carbon dioxide, _____, and _____.
    a. Oxygen; water
    b. Oxygen; glucose
    c. Water; ATP
    d. Carbon monoxide; glucose

10. The purpose of a biology lab was to measure the oxygen consumption during seed germination as evidence of the rate of cellular respiration. Both germinating and non-germinating peas at different temperatures were used. Glass beads placed in one respirometer served as the control. Considering the chemical reaction for cellular respiration, what must be included in the respirometer containing the germinating seeds?
    a. Carbon dioxide
    b. Water
    c. Oxygen
    d. Light

Station 8: PS Practice Questions

1. Which is not a way in which life is dependent on photosynthesis?
    a. Production of oxygen
    b. Synthesis of sugar
    c. Recycling of carbon dioxide
    d. Converting oxygen and hydrogen into water

2. Which pigment molecule absorbs blue and red light to provide the energy needed for photosynthesis?
    a. Carotenoid
    b. Chlorophyll
    c. Melanin
    d. Anthocyanin

3. Which energy carrying molecule is made during the light reactions to be used during the Calvin cycle?
    a. H2O
    b. NADH
    c. FADH2
    d. ATP

4. Which structures facilitate the movement of gasses in and out of plant cells?
    a. Plasmodesmata
    b. Stroma
    c. Stomata
    d. Nuclear pore

5. A student is setting up an experiment looking at the rate of photosynthesis in pea plants. Which of the following factors would NOT affect the student's experiment?
    a. Light intensity
    b. Temperature
    c. Oxygen concentration
    d. Carbon dioxide concentration

6. Four identical plants are grown under different colored light bulbs. Under which color will the release of oxygen be the slowest?
    a. Blue
    b. Orange
    c. Green
    d. Red

7. In photosynthesis, there are two subunits (photosystem I and photosystem II). These systems occur in the _____ membranes of the chloroplast. Photosystem I (PS I) and photosystem II (PS II) are _____ reactions.
    a. stroma; light-dependent
    b. stroma; light-independent
    c. thylakoid; light-dependent
    d. thylakoid; light-independent

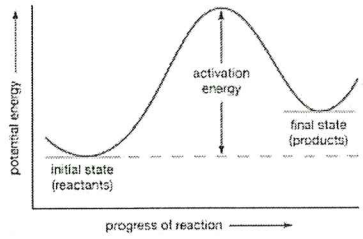
8. Photosynthesis is composed of two main parts (light-dependent and light-independent). Photosystem I (PS I) and photosystem II (PS II) are alike in many ways. How does photosystem I and photosystem II differ from one another?
    a. PS I produces glucose while PS II produces ATP.
    b. PS I occurs in the stroma while PS II takes place in the thylakoid membranes.
    c. PS I involves NADP+ Reductase while PS II involves ATP Synthase
    d. PS I splits water while PS II makes Oxygen

9. Photosynthesis has two main parts (light-independent and light dependent). Which descriptions can BEST be used to explain the calvin cycle?

    I. Light-independent reaction
    II. Light-dependent reaction
    III. Products are ATP and NADPH.
    IV. Occurs in the thylakoid membranes
    V. Occurs in the stroma
    VI. Glucose is made from CO_2

    a. I, V, and VI
    b. II, III, and IV
    c. I, IV, V, and VI
    d. I, II, III, and V

10. The graph here shows the relative change in energy level for the reactants and the products of photosynthesis. Given the energy levels modeled in the graph, which statement accurately compares the energy of the products and the reactants with regards to photosynthesis?



a. Water and oxygen have the highest energy compared to glucose and carbon dioxide.

b. Glucose and oxygen gas have higher energy levels compared to water and carbon dioxide.

c. Glucose and water have the same energy levels compared to carbon dioxide and oxygen gas.

d. Glucose and oxygen have a lower energy level associated with them compared to carbon dioxide and water.

Extra Practice Questions

1. What are the three parts of ATP and where is the majority of energy stored in ATP?

2. What is an autotroph? What is a heterotroph?

3. Which organisms go through cellular respiration? Photosynthesis?

4. Why does fermentation happen? Where does it happen?

5. **Compare** and **contrast** both types of fermentation.

6. What does fermentation **recycle**? Where does this go and why is it significant?

7. Lactic acid fermentation is reversible but alcoholic fermentation is not. Why?

8. What is the SPECIFIC product of the Calvin Cycle and what is it used to make?

9. What is cycled back into the krebs cycle?

10. What is cycled back into the Calvin cycle?

Write the chemical equation for each process above each diagram, then label everything.

